

High temperatures predict altered gut microbiome composition, metabolic capacity, and immune interactions in a wild primate species

Running title: Heat alters primate gut microbiomes

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Abstract: Rising temperatures due to climate change are impacting the function of critical physiological processes in wild animals. Heat stress may compromise the gut microbiome's ability to support essential host functions, highlighting an important pathway by which climate change could negatively affect numerous species. However, the relationship between the microbiome and heat stress has been studied primarily in experimental contexts and remains poorly understood in ecologically relevant environments. Here, we examine how natural thermal variation influences gut microbiome taxonomic composition and functional capacity in a wild population of white-face capuchin monkeys. We find that high temperatures are associated with altered microbiome community-level diversity, taxonomic composition, and metabolic capacity consistent with physiological responses to heat stress. We also show that the microbiome's response to high temperatures is linked to host immunity and may affect microbiome-immune interactions. These results suggest that rising temperatures may compromise essential microbiome functions and highlight a potential mechanism by which climate change could negatively impact species through their microbiomes.